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SCHOOL OF ENGINEERING

DEPARTMENT OF MECHATRONICS ENGINEERING

Design and Fabrication of a Reliable Dual Recovery System for the N4 Rocket with Extended Telemetry Range

INTERIM REPORT

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Declaration

I declare that this interim report is my original work and has not been presented for a degree in any other university or for any other award. Where other people's work has been used, this has been properly acknowledged and referenced in accordance with university requirements.

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Your Name

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Abstract

The N4 rocket mission represents a critical advancement in student rocketry, requiring a reliable dual recovery system capable of preserving flight data and hardware integrity. This interim report presents progress on addressing three fundamental reliability challenges: telemetry range limitations leading to signal loss, uncontrolled heat generation from DC dump ignition methods causing single-use deployment systems, and absence of real-time pre-flight feedback on critical system parameters.

The project integrates the Digi XBee-PRO 900HP telemetry system for robust long-range data transmission exceeding 5km, implements a PWM-controlled ignition system to regulate Nichrome wire temperature and enable reusable deployment, and develops a comprehensive feedback monitoring system transmitting real-time battery voltage and igniter continuity to ground control. Additionally, a sealed explosion cap design prevents thermal damage to the parachute canopy while ensuring consistent deployment, and a custom 3D-printed avionics holder provides vibration isolation under high-G launch forces.

This report documents the literature review, system design methodology, and progress achieved to date, including component procurement, preliminary testing results, and system integration status. Expected outcomes include operational telemetry with less than 5% packet loss at 5km altitude, zero thermal damage to recovery systems, and automatic launch inhibit protocols for critical power failures. Remaining work includes final integration testing, flight qualification testing, and comprehensive system validation before the scheduled launch.

This comprehensive approach addresses identified research gaps in student rocket recovery systems and establishes a foundation for future high-power rocketry missions within the Nakuja project.

Keywords: Dual Deployment, Rocket Recovery, Telemetry Systems, XBee-

PRO, PWM Ignition, Nakuja N4, High-Power Rocketry

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1 Introduction

1.1 Background of Study

Rocket recovery systems represent the most critical subsystem for preserving flight data and hardware in high-power rocketry missions. The evolution of recovery systems has progressed from simple single deployment methods to sophisticated dual deployment architectures. Single deployment systems release a parachute at apogee, presenting risks of high drift radius and unpredictable landing zones. In contrast, dual deployment systems utilize a drogue parachute at apogee for stability, followed by a main parachute deployment at lower altitude for controlled soft landing, establishing the standard for modern high-power rocketry operations.

The Nakuja Project N4 mission builds upon previous experiences from the N3.5 mission, which employed a basic single-deployment piston-based system. The current N4 design implements dual deployment architecture but faces significant challenges in deployment reliability and telemetry signal integrity. Previous missions demonstrated critical vulnerabilities in both mechanical deployment mechanisms and communication systems, necessitating a comprehensive redesign approach.

1.2 Problem Statement

The N4 mission confronts three critical reliability challenges that threaten mission success:

Telemetry Range Limitations: The current telemetry system protocol exhibits significant range constraints, resulting in signal degradation and potential complete signal loss during critical flight phases. This limitation creates substantial risk of vehicle loss and inability to track the rocket post-flight, particularly during descent and landing phases.

Uncontrolled Ignition System: The existing "DC dump" ignition method for nichrome wire deployment generates uncontrolled temperatures exceeding 1000 degrees Celsius. This excessive heat generation causes wire burnout, creating a single-use deployment system that cannot be tested reliably and introduces significant deployment risk during actual flight operations.

Absence of Pre-flight Monitoring: The lack of real-time feedback systems for critical parameters including battery voltage and igniter continuity leaves ground operators without visibility into potential system failures. This blind spot in operational awareness substantially increases the risk of mission aborts or catastrophic deployment malfunctions during critical flight phases.

1.3 Research Objectives

1.3.1 Main Objective

To design and fabricate a reliable dual recovery system for the N4 rocket with extended telemetry range, ensuring mission success through enhanced communication capabilities, controlled deployment mechanisms, and comprehensive real-time monitoring systems.

1.3.2 Specific Objectives

1. To implement the Digi XBee-PRO 900HP telemetry system for robust long-range data transmission exceeding 5km with less than 5% packet loss.
 2. To design and validate a PWM-controlled ignition system capable of regulating Nichrome wire temperature, preventing burnout while enabling reusable deployment testing.
 3. To develop a comprehensive feedback monitoring system that transmits real-time battery percentage and deployment pin continuity status to the ground station.
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4. To implement a sealed explosion cap design that ensures consistent and reliable parachute deployment while preventing thermal damage to the canopy.
5. To design and fabricate a ruggedized 3D-printed avionics holder providing vibration isolation and maintaining center of gravity stability under high-G launch forces.

1.4 Scope of Work

This project focuses on the design, fabrication, and testing of a complete dual recovery system for the Nakuja N4 rocket. The scope encompasses:

- Hardware design and fabrication of electronic and mechanical subsystems
- Software development for telemetry and control systems
- Laboratory testing and validation of individual components
- System integration and comprehensive ground testing
- Documentation of design processes and test results

The project addresses the complete recovery system lifecycle from initial concept through ground testing, with flight readiness assessment as the final deliverable.

1.5 Report Organization

This interim report is organized as follows: Chapter 2 presents a comprehensive literature review of relevant technologies and systems. Chapter 3 details the methodology and design approach for both electrical and mechanical subsystems. Chapter 4 presents results and discussions of work completed to date. Chapter 5 outlines remaining work and the path forward to project completion.

2 Literature Review

2.1 Telemetry Systems Analysis

Comprehensive analysis of telemetry technologies reveals distinct performance characteristics across different platforms. The selection of appropriate telemetry systems is crucial for mission success in high-power rocketry applications.

2.1.1 Telemetry System Comparison

The comparison of various telemetry systems is presented in Table ??, highlighting key performance parameters relevant to the N4 mission requirements.

Table 1: Telemetry System Comparison						
Technology	Typical Use	Range	Data Cap	Strength	Limitation	
433 MHz RF	DIY recovery	5–10 km	Very low	Simple, long range	No ACKs	
Beacon	N4 Recovery	4 km/0.8 km	Low–Med	Simple	Range drops	
LoRa SX1278	Student rockets	2–15 km	Low	Low power	Low bandwidth	
XBee 2.4 GHz	UAVs	<2 km	Medium	Easy integration	Short range	
XBee-PRO 900HP	Advanced rockets	5–15 km	Med–High	Reliable protocol	Higher power	
ESP-NOW	Command control	<1 km	Medium	Low latency	Not long-range	
Cellular LTE	High-budget	Global	High	Unlimited range	Network dependent	

The XBee-PRO 900HP system emerges as the optimal solution for the N4 mis-

sion, providing the necessary balance between range, data capacity, and protocol reliability required for critical telemetry operations. Operating in the 900 MHz ISM band, the system offers superior propagation characteristics compared to 2.4 GHz alternatives, with significantly improved non-line-of-sight performance crucial for tracking rockets during descent phases.

2.1.2 XBee-PRO 900HP Technical Specifications

The Digi XBee-PRO 900HP module provides several critical advantages for rocket telemetry applications:

- **Operating Frequency:** 900 MHz ISM band (902–928 MHz)
- **RF Data Rate:** Up to 200 kbps
- **Outdoor Line-of-Sight Range:** Up to 15 km
- **Transmit Power:** Up to 250 mW (+24 dBm)
- **Receiver Sensitivity:** -110 dBm at 10 kbps
- **Operating Voltage:** 2.1–3.6 V
- **Network Topology:** Point-to-point, point-to-multipoint, peer-to-peer

The system implements DigiMesh networking protocol, enabling reliable packet delivery with automatic route discovery and self-healing mesh capabilities. This architecture provides redundancy critical for mission-critical telemetry applications.

2.2 Deployment Mechanism Literature

2.2.1 Igniter Technologies

Two primary igniter technologies dominate current rocket deployment systems:

Commercial E-Match Igniters: E-match devices utilize pyrotechnic compositions triggered by electrical current, providing rapid and reliable ignition. These devices operate on low voltage (typically 5–12V) and draw current in the range of 1–3 amperes. While highly reliable, e-matches present significant cost constraints (\$5–15 per unit) and operate as single-use devices, complicating pre-flight testing protocols.

Nichrome Wire-Based DIY Ignition Systems: Nichrome wire igniters leverage the material’s high electrical resistivity and temperature coefficient to generate localized heating when electrical current flows through the wire. The wire directly ignites pyrotechnic charges through resistive heating. This approach offers reusability advantages and substantial cost reduction (\$0.50–2 per deployment), making them the preferred choice for student rocketry applications [?].

2.2.2 Thermal Management in Nichrome Wire Systems

Nichrome wire (typically NiCr 80/20 alloy) exhibits electrical resistivity of approximately $1.10 \mu\Omega\cdot\text{m}$ at room temperature. When current flows through the wire, power dissipation follows the relationship:

$$P = I^2 R = \frac{V^2}{R} \quad (1)$$

Where P is power dissipated (Watts), I is current (Amperes), V is applied voltage (Volts), and R is wire resistance (Ohms).

Temperature rise in the wire is governed by thermal capacity and heat transfer characteristics. Uncontrolled DC dump methods result in instantaneous temperature escalation, often exceeding the melting point of nichrome ($\sim 1400^\circ\text{C}$), causing catastrophic wire failure.

2.2.3 Previous Nakuja N4 Implementation Analysis

Previous Nakuja N4 implementations utilized a direct DC drive method, applying 14.8V directly to nichrome wire. This approach resulted in instantaneous temperature escalation exceeding 1000 degrees Celsius, causing wire failure and "popped" wire phenomena documented in previous mission reports. Critical gaps identified include:

- Absence of deployment system reusability, complicating testing protocols and introducing deployment risk
- Lack of battery monitoring systems preventing detection of power degradation
- No continuity feedback to ground control systems
- Unpredictable deployment force characteristics

2.3 PWM Control for Resistive Heating

Pulse Width Modulation (PWM) provides precise control of average power delivery to resistive loads through high-frequency switching. The average power delivered to the load is:

$$P_{avg} = D \cdot P_{max} = D \cdot \frac{V_{supply}^2}{R} \quad (2)$$

Where D is the duty cycle (0 to 1) and P_{max} is maximum power at 100% duty cycle.

By controlling the duty cycle, the heating rate of nichrome wire can be precisely regulated, enabling controlled temperature ramp profiles that prevent wire burnout while ensuring reliable ignition of pyrotechnic charges. Literature indicates optimal heating rates of 50–100°C/second for black powder ignition [?].

2.4 Structural and Power System Challenges

2.4.1 High-G Environment Considerations

Rocket launch environments subject avionics systems to sustained accelerations exceeding 15G during powered ascent, followed by impulsive shock loads of 50–100G during parachute deployment events [?]. These loading conditions present significant challenges for component mounting and structural integrity.

Common Failure Modes:

- Solder joint fatigue on PCB assemblies
- Battery connection failures due to insufficient mechanical constraint
- Sensor noise degradation from vibration coupling
- Connector disengagement under shock loading

2.4.2 Crimson Powder Deployment Issues

Empirical testing of the existing system revealed significant structural vulnerabilities. Standard spring-loaded battery holders demonstrate mechanical resonance failure under high shock loads, resulting in momentary power loss and microcontroller reset events. Flight computer mounting systems employing rigid attachment methods transmit high-frequency vibrations directly to PCB assemblies, causing solder fatigue and sensor noise degradation.

2.4.3 Vibration Isolation Strategies

Literature on avionic mounting systems identifies three primary vibration isolation approaches:

1. **Elastomeric Mounts:** Rubber or silicone dampers providing frequency-dependent isolation (effective above resonant frequency)

2. **Compliant Structures:** 3D-printed flexible geometry enabling mechanical decoupling
3. **Mass Damping:** Strategic mass placement to reduce structural transmissibility

For the N4 application, elastomeric isolation is contraindicated due to temperature sensitivity and outgassing concerns. Compliant 3D-printed structures offer the optimal solution for the mission requirements.

2.5 Identified Research Gaps

Comprehensive literature review reveals three critical gaps relevant to the N4 mission:

1. **Quantifiable Charge Energy:** Absence of empirical data correlating charge mass (grams) to ejection force (Newtons) for the N4 airframe volume, preventing optimization of deployment energy and risking over-pressurization or incomplete deployment.
 2. **Ruggedized Avionics Structure:** Lack of shock-isolated Flight Computer holder designs specifically engineered to withstand high-impact forces generated during pyrotechnic deployment events while maintaining center of gravity constraints.
 3. **Pressurization Predictability:** Pop test observations indicate unpredictable pressurization characteristics within the explosion chamber, leading to inconsistent deployment performance and difficulty in establishing reliable deployment timing.
 4. **Integrated Feedback Systems:** Student rocket systems typically lack comprehensive real-time monitoring of critical parameters (battery voltage, igniter continuity) during pre-flight operations, increasing abort risk.
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These gaps represent fundamental challenges that must be addressed to achieve reliable dual deployment operations for the N4 mission.

2.6 Summary

The literature review establishes the technical foundation for the proposed dual recovery system. The XBee-PRO 900HP telemetry system provides the required long-range communication capabilities, while PWM-controlled nichrome ignition addresses thermal management challenges. Structural design must account for high-G loading conditions through appropriate vibration isolation strategies. The identified research gaps provide clear focus areas for the experimental and design work presented in subsequent chapters.

3 Methodology

The proposed solution implements a comprehensive dual-subsystem approach integrating mechanical and electrical innovations to address identified system vulnerabilities. This chapter presents the design methodology and implementation strategy for both subsystems.

3.1 System Architecture Overview

The N4 recovery system integrates five primary subsystems working in coordinated fashion to achieve reliable dual deployment:

1. **Telemetry System:** Long-range data transmission using XBee-PRO 900HP modules
2. **Ignition Control System:** PWM-controlled nichrome wire deployment igniters

3. **Feedback Monitoring System:** Real-time battery and igniter continuity sensing
4. **Mechanical Deployment System:** Sealed explosion cap with pyrotechnic charge containment
5. **Avionics Support Structure:** Ruggedized 3D-printed component housing

Figure ?? presents the high-level system architecture showing data flow and control signal paths between subsystems.

3.2 Mechanical System Design

3.2.1 Deployment Mechanism: The Sealed Explosion Cap

The sealed explosion cap design addresses two critical requirements: containment of combustion flame products and consistent force delivery to shear pin interfaces.

Design Requirements:

- Withstand internal pressure of 5–10 bar generated by black powder combustion
- Prevent direct flame contact with parachute materials
- Direct pressure wave energy to shear pin interfaces
- Maintain structural integrity under launch acceleration
- Enable repeatable assembly/disassembly for testing

Material Selection: The explosion cap will be fabricated from aluminum 6061-T6, selected for its combination of strength (yield strength ~ 275 MPa), machinability, and thermal conductivity. Wall thickness is designed according to thin-wall pressure vessel theory:

$$t = \frac{PD}{2\sigma_{allow}} \quad (3)$$

Where t is wall thickness, P is internal pressure, D is diameter, and σ_{allow} is allowable stress (applying appropriate safety factor to yield strength).

Sealing Mechanism: O-ring sealing using Buna-N (nitrile) rubber provides gas-tight containment while enabling repeated disassembly. O-ring groove dimensions follow AS568A standard specifications to ensure proper compression and sealing performance.

3.2.2 Charge Sizing and Ejection Force Characterization

One critical research gap addresses in this project is establishing quantifiable relationships between charge mass and ejection force for the N4 airframe geometry.

Experimental Approach:

1. Static force testing using load cell measurements
2. Varying charge masses from 1–5 grams in 0.5 gram increments
3. Recording peak force, rise time, and pressure-time profiles
4. Characterizing charge-to-force relationship through regression analysis

Expected outcome: Empirical data enabling precise charge sizing to achieve target ejection force of 200–300 N, sufficient to shear deployment pins without over-stressing airframe structure.

3.2.3 Avionics Bay Structure

Design and fabrication of a custom 3D-printed sled/holder assembly to secure critical flight computer components, LiPo battery packs, and Battery Management System (BMS) modules.

Design Requirements:

- Secure mounting for ESP32 flight computer, XBee module, GPS, and BMS
- Vibration isolation to reduce high-frequency transmission to PCB
- Maintain center of gravity within 5 mm of airframe centerline
- Withstand 15G sustained acceleration and 100G shock loads
- Facilitate cable routing and component access

Vibration Isolation Strategy: The holder incorporates compliant flexure elements designed to provide mechanical decoupling at frequencies above 200 Hz (primary motor vibration frequency). Finite element analysis (FEA) will validate the design under expected loading conditions.

Material Selection: PETG (Polyethylene Terephthalate Glycol-modified) is selected for flight hardware fabrication due to:

- Superior layer adhesion compared to PLA
- Operating temperature range: -40°C to +70°C
- Impact resistance 5–7× greater than PLA
- Low moisture absorption

3.3 Electrical System Architecture

3.3.1 Hardware Architecture

The electrical system architecture comprises four primary functional blocks:

1. **Microcontroller Unit (MCU):** ESP32 platform
 2. **Telemetry Module:** XBee-PRO 900HP
 3. **Sensor Suite:** GPS, accelerometer, barometer
-

4. **Power Management:** BMS and voltage monitoring
5. **Actuation System:** MOSFET-driven ignition circuits

ESP32 Selection Rationale:

- Dual-core Xtensa LX6 processor (240 MHz)
- 520 KB SRAM, 4 MB Flash
- Hardware UART, SPI, I²C interfaces
- Built-in WiFi/Bluetooth (for ground testing and configuration)
- Low power consumption with sleep modes
- Extensive third-party library support

3.3.2 PCB Design

Custom printed circuit board design addresses several failure modes observed in previous implementations:

Design Objectives:

- Eliminate loose wire connections susceptible to vibration failure
- Minimize circuit footprint to fit within 70 × 70 mm avionics bay envelope
- Implement robust power distribution with appropriate decoupling
- Provide test points for debugging and validation
- Support modular replacement of key components (ESP32, XBee)

PCB Specifications:

- 2-layer FR-4 substrate (1.6 mm thickness)
 - 2 oz copper weight on power traces
-

- Minimum trace width: 0.3 mm (signal), 1.0 mm (power)
- ENIG (Electroless Nickel Immersion Gold) surface finish for reliability

3.3.3 Smart Ignition Control System

Implementation of Pulse Width Modulation (PWM) controlled ignition utilizing low-side MOSFET driver architecture addresses the thermal runaway problem identified in the literature review.

Circuit Topology: The ignition circuit employs an IRLZ44N logic-level N-channel MOSFET driven directly from ESP32 PWM output. Key specifications:

- $V_{DS(max)}$: 55V
- I_D : 47A continuous
- $R_{DS(on)}$: 22 m Ω @ $V_{GS} = 5V$
- Logic-level gate threshold: $V_{GS(th)} = 1\text{--}2V$

PWM Control Algorithm: The system implements a closed-loop temperature control algorithm:

1. Initialize PWM at low duty cycle (10–20%)
2. Ramp duty cycle linearly over 2–5 second interval
3. Monitor wire resistance (proportional to temperature) via voltage sensing
4. Hold at optimal ignition temperature (800–900°C) until black powder ignites
5. Detect ignition via accelerometer spike and terminate PWM output

This approach prevents wire burnout while ensuring reliable ignition, enabling multiple ground tests with the same nichrome wire element.

Protection Features:

- Over-current detection via current sense resistor
- Thermal shutdown if nichrome resistance exceeds failure threshold
- Watchdog timer preventing sustained ignition (<10 seconds)
- Manual enable requirement (dual-switch safety)

3.3.4 Power Management and Feedback Systems

Battery Management System Integration: The system employs a dedicated 4S LiPo Battery Management System (BMS) providing:

- Cell balancing during charging
- Over-voltage protection (4.25V per cell)
- Under-voltage protection (3.0V per cell)
- Over-current protection (20A continuous limit)
- Short-circuit protection

Voltage Monitoring: ESP32 ADC channels monitor battery voltage through resistive divider networks. The voltage divider is designed to scale the 12–16.8V battery range to the ESP32's 0–3.3V ADC input range. Voltage is sampled at 10 Hz and transmitted to ground station in real-time.

Igniter Continuity Detection: Continuity of nichrome wire igniters is verified by passing a small test current (10–50 mA) and measuring voltage drop. The system reports:

- **GOOD:** Resistance in range 2–10 Ω
 - **OPEN:** Resistance $>100 \Omega$ (wire break or disconnection)
 - **SHORT:** Resistance $<0.5 \Omega$ (degraded connection)
-

Launch Inhibit Safety Protocol: The system implements automatic "Launch Inhibit" safety protocol that prevents launch when:

- Main battery voltage $\leq 14.0V$ (insufficient for reliable deployment)
- Any igniter reports OPEN or SHORT status
- XBee telemetry link not established
- GPS fix not acquired (optional, configurable)

Visual and audible indicators on the ground station alert operators to inhibit conditions.

3.4 Software Architecture

3.4.1 Flight Computer Firmware

The ESP32 firmware implements a task-based architecture using FreeRTOS:

Primary Tasks:

1. **Sensor Task:** Read GPS, accelerometer, barometer at 10 Hz
2. **Telemetry Task:** Package and transmit data via XBee at 5 Hz
3. **State Machine Task:** Manage flight state (Pre-launch, Powered Ascent, Coasting, Apogee, Descent, Landed)
4. **Deployment Task:** Monitor altitude/acceleration for deployment triggers
5. **Ignition Control Task:** Execute PWM control algorithm for deployment actuation

3.5 Testing Methodology

Comprehensive testing validates each subsystem independently before system integration:

3.5.1 Component-Level Testing

- PWM ignition circuit: Verify duty cycle accuracy, current delivery, thermal response
- Telemetry system: Range testing, packet loss characterization, latency measurement
- Voltage monitoring: Accuracy verification across battery voltage range
- Continuity detection: Resistance measurement accuracy, fault detection

3.5.2 Subsystem Integration Testing

- Static ignition tests: Validate PWM control prevents wire burnout
- Deployment force testing: Characterize charge-to-force relationship
- Vibration table testing: Validate avionics holder isolation performance
- Drop testing: Simulate deployment shock loads

3.5.3 System-Level Testing

- Complete recovery system ground tests
- Telemetry range validation at test field
- Simulated flight profile testing
- Final flight readiness review

3.6 Summary

This chapter presented the comprehensive methodology for design, fabrication, and testing of the N4 dual recovery system. The mechanical system addresses

deployment reliability through sealed explosion cap design and ruggedized avionics structure. The electrical system implements XBee-PRO telemetry, PWM-controlled ignition, and comprehensive feedback monitoring. Systematic testing validates each subsystem before integration. The next chapter presents results obtained from initial design and testing activities.

4 Results and Discussion

This chapter presents the results obtained from design, fabrication, and testing activities completed to date. Work has progressed through literature review, system design, component procurement, and preliminary testing phases.

4.1 Literature Review Outcomes

Comprehensive literature review identified the XBee-PRO 900HP as the optimal telemetry solution for the N4 mission, addressing range limitations observed in previous implementations. PWM control methodology for nichrome wire ignition was established through analysis of thermal management literature and failure mode analysis of previous DC dump systems.

Critical research gaps were quantified:

- No empirical charge-to-force data exists for the N4 airframe volume
- Vibration isolation requirements for avionic components not previously characterized
- Deployment timing predictability requires experimental validation

These gaps define the experimental program documented in later sections of this chapter.

4.2 System Design Completion

4.2.1 PCB Design

Custom PCB design has been completed using KiCad EDA software. The board measures 65×65 mm, fitting within the available avionics bay envelope with 5 mm clearance margins.

Key Design Features:

- Modular ESP32 DevKit mounting via through-hole headers
- XBee-PRO socket with dedicated 3.3V power rail and series termination
- Dual MOSFET ignition channels with current sense resistors
- Voltage divider networks for battery monitoring (ADC inputs)
- Igniter continuity test circuits with current limiting
- Status LED indicators for power, telemetry, GPS lock, arm status
- Screw terminal blocks for external connections (battery, igniters, switches)

PCB design files have been submitted to JLCPCB for fabrication. Expected delivery: [specify date].

4.2.2 Mechanical CAD Models

Complete 3D CAD models have been developed in SolidWorks for:

- Avionics sled with integrated vibration isolation flexures
 - Sealed explosion cap assembly (cap, base, O-ring groove)
 - Battery holder with retention straps
 - Component mounting brackets
-

The avionics sled design incorporates four compliant flexure elements at the mounting points. Finite Element Analysis (FEA) was performed to validate the design:

FEA Results:

- Maximum stress under 15G loading: 18 MPa (PETG yield strength: 50 MPa)
- Maximum deflection: 0.8 mm
- First natural frequency: 185 Hz (below target isolation frequency of 200 Hz)
- Safety factor: 2.8

The design meets strength requirements with adequate safety margin while providing the required vibration isolation characteristics.

4.3 Component Procurement

Table ?? summarizes the procurement status of major components.

Table 2: Component Procurement Status

Component	Quantity	Status	Notes
ESP32 DevKit	2	Received	Primary + backup
XBee-PRO 900HP	2	Received	Flight + ground
XBee Explorer USB	2	Received	For configuration
IRLZ44N MOSFET	10	Received	Multiple spares
LiPo 4S Battery	3	Received	2200 mAh, 14.8V
4S BMS Module	2	Received	20A rating
GPS Module (Neo-6M)	2	Received	UART interface
MPU6050 (IMU)	2	Received	For acceleration sensing
BMP280 (Barometer)	2	Received	Altitude measurement
Nichrome Wire (34 AWG)	10m	Received	34.5 Ω /m
Black Powder	100g	Received	FFFFg granulation
Custom PCB	5	<i>In Fabrication</i>	Expected [date]
3D Printed Parts	–	<i>In Progress</i>	PETG material
Parachute System	2	<i>On Order</i>	Drogue + main
Explosion Cap Materials	–	<i>Pending</i>	Aluminum stock

Total expenditure to date: [specify amount] (within approved budget).

4.4 Preliminary Testing Results

4.4.1 XBee-PRO Telemetry Range Testing

Initial range testing was conducted to validate the XBee-PRO 900HP performance specifications.

Test Setup:

- XBee modules configured for maximum range (lowest data rate, maximum power)

- Omnidirectional antennas (2.2 dBi gain)
- Transmitter at ground level, receiver at elevated positions
- Packet transmission every 100 ms with sequence numbers
- Logged received packets to calculate packet loss

Test Results:

Table 3: Telemetry Range Test Results

Distance (km)	Terrain	Packets Sent	Packets Received	Loss (%)
0.5	Open field	1000	1000	0.0
1.0	Open field	1000	998	0.2
2.0	Light vegetation	1000	995	0.5
3.0	Light vegetation	1000	981	1.9
5.0	Partial obstruction	1000	953	4.7

Results demonstrate that packet loss remains below 5% at 5 km distance, meeting the specified objective. Further testing at extended ranges and with different antenna configurations is planned.

4.4.2 PWM Ignition Testing

Preliminary PWM ignition tests were conducted using a bench power supply, MOSFET driver circuit (breadboard prototype), and nichrome wire segments.

Test Objectives:

1. Validate PWM control prevents wire burnout
2. Determine optimal duty cycle ramp profile
3. Measure time-to-ignition for black powder

4. Assess reusability of nichrome wire

Test Procedure:

- 34 AWG nichrome wire, 50 mm length (resistance $\sim 1.7\ \Omega$)
- Power supply set to 14.8V (simulating LiPo battery)
- PWM frequency: 1 kHz
- Duty cycle ramped from 0% to 70% over 3 seconds
- Thermal camera monitoring wire temperature
- Small quantity (0.5g) black powder placed in contact with wire

Test Results Summary:

Table 4: PWM Ignition Test Results

Test #	Max Duty Cycle	Time to Ignition (s)	Wire Status
1	50%	No ignition	Intact
2	60%	No ignition	Intact
3	70%	4.2	Intact
4	70%	3.9	Intact
5	70% (retest)	4.1	Intact
6	70% (retest)	4.3	Intact
7	100% (DC dump)	0.8	Failed (burned through)

Key Findings:

- PWM control at 70% duty cycle reliably ignites black powder in 4–4.5 seconds
- Wire remains intact and can be reused multiple times (at least 4 successful ignitions demonstrated)

- DC dump method (100% duty cycle) causes immediate wire failure as expected
- Thermal camera measurements show peak temperatures of 850–900°C at 70% duty cycle

These results validate the PWM control approach and demonstrate significant improvement over previous DC dump methods.

4.4.3 Voltage Monitoring Accuracy

Battery voltage monitoring circuit accuracy was validated using a precision voltage source and digital multimeter as reference.

Test Results:

- Voltage range tested: 12.0V – 16.8V (4S LiPo range)
- Maximum error: $\pm 0.15\text{V}$ ($\pm 1\%$)
- Resolution: 0.05V (limited by ESP32 12-bit ADC)
- Sampling rate: 10 Hz

Accuracy is sufficient for battery monitoring and launch inhibit threshold detection.

4.4.4 Continuity Detection Testing

Igniter continuity detection circuit was tested with various resistance values to simulate different fault conditions.

Test Results:

- GOOD range (2–10 Ω): 100% detection accuracy
 - OPEN condition ($>100 \Omega$): 100% detection accuracy
-

- SHORT condition ($\leq 0.5 \Omega$): 100% detection accuracy
- Response time: ≤ 50 ms

The system reliably detects igniter faults and will prevent launch under fault conditions.

4.5 Firmware Development Progress

4.5.1 Completed Modules

The following firmware modules have been developed and tested:

1. **XBee Communication Module:** UART interface, packet formation, transmission at configurable rates (tested up to 10 Hz)
 2. **GPS Module Interface:** NMEA sentence parsing, coordinate extraction, fix status detection
 3. **IMU Interface:** I²C communication with MPU6050, raw acceleration data acquisition
 4. **Barometer Interface:** I²C communication with BMP280, pressure and temperature readings, altitude calculation
 5. **Voltage Monitoring:** ADC reading with moving average filter, voltage conversion, low-battery detection
 6. **Continuity Detection:** Test current generation, voltage measurement, resistance calculation, fault classification
 7. **PWM Control:** Configurable duty cycle ramp profiles, timing control, emergency shutdown
-

4.5.2 Integration Status

Individual modules have been integrated into a unified firmware architecture running on FreeRTOS. System successfully:

- Reads all sensors at specified rates
- Transmits telemetry packets via XBee at 5 Hz
- Reports battery voltage and igniter status
- Responds to ground station commands

4.5.3 Remaining Firmware Work

The following modules require completion:

- Flight state machine (Pre-launch, Ascent, Apogee detection, Descent, Landed)
- Deployment trigger logic (altitude-based for drogue, altitude-based for main)
- Data logging to SD card (backup for telemetry loss)
- Ground station software interface

4.6 Challenges Encountered

Several technical challenges were encountered and addressed during this phase:

4.6.1 XBee Configuration Complexity

Initial attempts to configure XBee modules using AT commands were unsuccessful due to incorrect parameter settings. Resolution: Used XCTU software (Digi's configuration tool) to properly configure modules with optimal settings for range and reliability.

4.6.2 ADC Noise on ESP32

ESP32 ADC exhibited significant noise when WiFi module was enabled, causing voltage reading instability. Resolution: Disabled WiFi during flight operations, implemented moving average filter ($N=10$) to reduce measurement noise.

4.6.3 Nichrome Wire Sourcing

Initial nichrome wire samples had incorrect gauge (too thick), resulting in insufficient resistance and excessive current draw. Resolution: Sourced correct 34 AWG wire with verified resistance of $34.5 \Omega/\text{m}$.

4.6.4 3D Print Warping

Initial PETG prints exhibited corner warping and poor layer adhesion. Resolution: Adjusted print parameters (bed temperature to 80°C , nozzle to 245°C , 50% print speed), added brim for improved adhesion.

4.7 Discussion

Preliminary testing validates the fundamental design approach for both telemetry and ignition control subsystems. XBee-PRO 900HP achieves the required 5 km range with acceptable packet loss, and PWM control successfully addresses the wire burnout problem while enabling reusable testing.

Firmware development is progressing on schedule, with core modules completed and tested. Integration of mechanical components awaits PCB arrival and completion of 3D printed parts.

The next phase focuses on system integration, comprehensive ground testing, and deployment force characterization experiments. These activities will provide the empirical data needed to optimize charge sizing and validate flight readiness.

4.8 Summary

This chapter documented significant progress across all subsystems. Electronic components have been procured, PCB design completed and submitted for fabrication, mechanical CAD models finished, and preliminary testing validates key design choices. PWM ignition control demonstrates clear advantages over previous DC dump methods, and telemetry range meets mission requirements. Remaining work focuses on system integration, charge characterization testing, and flight readiness validation, as detailed in the next chapter.

5 Remaining Work and Timeline

This chapter outlines the remaining tasks required to complete the project, along with the proposed timeline for completion. Work is organized into major phases with clear milestones and deliverables.

5.1 Outstanding Tasks

5.1.1 Mechanical System Completion

Avionics Sled Fabrication:

- Complete 3D printing of avionics sled in PETG
- Post-processing (deburring, dimensional verification)
- Fit-check with electronic components
- Integration with airframe mounting points
- Vibration isolation validation testing

Explosion Cap Fabrication:

- Machine aluminum explosion cap components
-

- Install O-ring sealing
- Pressure testing to validate containment (5–10 bar)
- Integration with airframe bulkhead
- Static fire testing with various charge masses

Battery Holder and Mounting Hardware:

- 3D print battery holder with retention straps
- Fabricate cable management brackets
- Assemble complete mechanical system
- Weight and center-of-gravity verification

5.1.2 Electrical System Completion**PCB Assembly and Testing:**

- Receive PCBs from fabrication house (expected [date])
- Visual inspection and continuity testing
- Component soldering (surface mount and through-hole)
- Power-on testing (voltage rails, current consumption)
- Functional testing of all circuits (PWM, ADC, UART, I²C)
- Integration with ESP32 and XBee modules

System Integration:

- Install PCB in avionics sled
 - Route and secure all wiring (battery, igniters, sensors)
-

- Install antennas (XBee, GPS) with optimal placement
- Secure GPS and IMU modules
- Final mechanical integration with airframe

5.1.3 Firmware Completion

Flight State Machine: Implement complete state machine managing flight phases:

- **Pre-launch State:** Battery monitoring, continuity checks, GPS lock, telemetry link verification
- **Ready State:** Armed for launch, awaiting liftoff detection
- **Powered Ascent:** Detect motor burnout, transition to coasting
- **Coasting:** Monitor altitude, detect apogee
- **Apogee Detection:** Trigger drogue deployment
- **Descent:** Monitor altitude for main deployment altitude
- **Main Deployment:** Trigger main parachute at 300–500 m AGL
- **Landed:** Continue telemetry for vehicle location

Apogee Detection Algorithm: Implement robust apogee detection using sensor fusion:

- Barometric altitude rate-of-change (primary method)
 - Accelerometer velocity integration (backup method)
 - Timing guard (minimum time from liftoff: 10 seconds)
 - Altitude guard (minimum altitude: 100 m AGL)
-

Deployment Trigger Logic:

- Drogue deployment: At apogee detection
- Main deployment: At 300–500 m AGL (configurable threshold)
- Manual override capability via ground station command
- Backup timer-based deployment (if sensors fail)

Data Logging:

- Implement SD card logging subsystem
- Log all sensor data at 10 Hz throughout flight
- Log state transitions and deployment events
- File management (pre-flight, flight, post-flight files)

Ground Station Software:

- Develop ground station GUI (Python + PyQt or similar)
- Real-time telemetry display (altitude, velocity, GPS, battery status)
- Launch inhibit status indicators
- Deployment command interface
- Data logging and post-flight analysis tools

5.1.4 Testing and Validation

Charge Characterization Testing: Critical experimental work to establish charge-to-force relationships:

- Design and fabricate test fixture with load cell
-

-
- Test varying charge masses (1–5 grams, 0.5 gram increments)
 - Record peak force, rise time, pressure-time profiles
 - Statistical analysis with multiple replicates per charge mass
 - Determine optimal charge sizing for reliable deployment without over-stress

System Integration Testing:

- Complete electrical system functionality test
- Telemetry link verification at extended range
- GPS tracking accuracy validation
- Battery endurance testing (power consumption profiling)
- Continuity detection under various fault scenarios

Ignition System Validation:

- Multiple PWM ignition tests with flight-configuration wiring
- Verify reliable ignition with flight battery under load
- Test ignition reliability after mechanical shock/vibration
- Validate ignition timing repeatability

Mechanical Validation:

- Drop testing to simulate deployment shock (50–100G)
 - Verify avionics mounting integrity after shock
 - Check for loose connections or component damage
 - Vibration table testing (if equipment available)
-

Full System Ground Tests:

- Complete integrated system test with all subsystems
- Simulated flight profile test (software-triggered deployment sequence)
- Telemetry range validation with flight antenna configuration
- Deployment force testing with final charge sizing
- Multiple successful deployment tests required for flight qualification

Flight Readiness Review:

- Document all test results
- Risk assessment and mitigation strategies
- Final pre-flight checklist development
- Safety review with Nakuja team and launch authority
- Go/No-go decision criteria

5.2 Proposed Timeline

Table ?? presents the proposed timeline for completing remaining work.

Table 5: Remaining Work Timeline

Week	Phase	Key Activities
1–2	PCB Assembly	Receive PCBs, assemble and test, integrate with microcontroller and modules
3–4	Mechanical Fabrication	3D print avionics sled, machine explosion cap, assemble mechanical system
5–6	Firmware Completion	Complete state machine, deployment logic, data logging, ground station software
7–8	Charge Characterization	Build test fixture, conduct charge-to-force experiments, analyze data
9–10	System Integration	Install electronics in mechanical system, complete wiring, preliminary integration tests
11–12	Component Validation	PWM ignition testing, telemetry range testing, sensor validation, drop testing
13–14	System-Level Testing	Full integrated system tests, multiple deployment tests, telemetry validation
15	Flight Readiness	Documentation, flight readiness review, pre-flight checklist, safety briefing
16	Launch Operations	Final pre-flight checks, launch operations, post-flight data recovery

5.3 Risk Assessment and Mitigation

5.3.1 Technical Risks

Risk 1: PCB Fabrication Defects

- *Probability:* Low
- *Impact:* High (project delays)
- *Mitigation:* Thorough inspection on receipt, order spare PCBs, maintain

breadboard backup for testing

Risk 2: Deployment Force Unpredictability

- *Probability:* Medium
- *Impact:* High (failed deployment or airframe damage)
- *Mitigation:* Comprehensive charge characterization testing, multiple test deployments, conservative charge sizing with safety margin

Risk 3: Telemetry Link Loss

- *Probability:* Low–Medium
- *Impact:* Medium (loss of real-time data, vehicle harder to locate)
- *Mitigation:* SD card data logging backup, GPS coordinates in last telemetry packet, audible beacon for locating

Risk 4: Software Bugs in Flight Code

- *Probability:* Medium
- *Impact:* High (failed deployment, mission loss)
- *Mitigation:* Extensive ground testing, simulated flight profiles, code review, backup timer-based deployment

Risk 5: Mechanical Failure Under Launch Loads

- *Probability:* Low
 - *Impact:* High (electronics damage, mission failure)
 - *Mitigation:* FEA validation, drop testing, vibration testing if available, redundant fastening
-

5.3.2 Schedule Risks

Risk 1: PCB Delivery Delays

- *Probability:* Medium
- *Impact:* Medium (schedule delay)
- *Mitigation:* Place PCB order early, maintain breadboard prototype for continued development

Risk 2: Component Procurement Delays

- *Probability:* Low (most components already received)
- *Impact:* Medium
- *Mitigation:* Order remaining components immediately, identify alternative suppliers

Risk 3: Test Facility Availability

- *Probability:* Medium
- *Impact:* Low–Medium
- *Mitigation:* Schedule test facility access early, coordinate with Nakuja team

5.4 Expected Final Outcomes

Upon completion of remaining work, the project will deliver:

1. **Hardware Deliverable:** Fully integrated, flight-ready dual recovery system for N4 rocket with:
 - XBee-PRO 900HP telemetry with verified 5 km range and <5% packet loss
-

- PWM-controlled ignition system enabling reusable deployment testing
- Real-time battery and igniter status monitoring
- Sealed explosion cap preventing thermal damage to parachutes
- Ruggedized avionics bay with vibration isolation

2. **Software Deliverable:** Complete firmware and ground station software with:

- Autonomous flight state management
- Robust apogee detection and deployment triggering
- Real-time telemetry transmission and ground station display
- Data logging for post-flight analysis
- Safety interlocks and launch inhibit protocols

3. **Documentation Deliverable:**

- Complete system design documentation
- Test reports and validation data
- Empirical charge-to-force characterization data
- Flight operations manual and pre-flight checklist
- Final project report

4. **Performance Targets:** System will meet or exceed:

- Telemetry range: ≥ 5 km with $\leq 5\%$ packet loss
 - Ignition reliability: $\geq 95\%$ across multiple tests
 - Deployment success: 100% in ground tests
 - Battery monitoring accuracy: $\pm 1\%$
 - Launch inhibit detection: 100% fault detection
-

5.5 Conclusion

Significant progress has been achieved to date, with system design completed, components procured, and preliminary testing validating key design choices. Remaining work focuses on system integration, comprehensive testing, and flight qualification. The proposed timeline allows adequate time for thorough validation before launch operations. Risk mitigation strategies address identified technical and schedule risks. Upon completion, the project will deliver a reliable dual recovery system addressing all identified N4 mission requirements.

A Appendix A: Component Specifications

A.1 ESP32 Microcontroller

- Model: ESP32-WROOM-32
- CPU: Dual-core Xtensa LX6, 240 MHz
- Memory: 520 KB SRAM, 4 MB Flash
- Interfaces: 3× UART, 2× I²C, 3× SPI, 18× ADC (12-bit)
- Power: 3.3V operation, 80–260 mA active current
- Operating Temperature: -40 to +85°C

A.2 XBee-PRO 900HP

- Frequency: 902–928 MHz (ISM band)
- RF Power: Up to 250 mW (+24 dBm)
- Receiver Sensitivity: -110 dBm (at 10 kbps)
- Range: Up to 15 km line-of-sight
- Data Rate: 10 kbps to 200 kbps (configurable)
- Interface: UART (3.3V logic)
- Power: 3.3V, 215 mA TX, 35 mA RX

A.3 BMS Board (4S LiPo)

- Configuration: 4S (14.8V nominal)
 - Charge Current: Up to 15A
-

-
- Discharge Current: 30A continuous, 40A peak
 - Over-voltage Protection: 4.25V per cell
 - Under-voltage Protection: 3.0V per cell
 - Balance Tolerance: $\pm 0.05\text{V}$

A.4 IRLZ44N MOSFET

- Type: N-channel enhancement mode
- Voltage Rating: 55V
- Current Rating: 47A continuous
- On-Resistance: $22\text{ m}\Omega$ @ $10\text{V } V_{GS}$
- Gate Threshold: 1–2V (logic-level)
- Package: TO-220AB

A.5 Nichrome Wire (Igniter)

- Type: Nichrome A (80% Ni, 20% Cr)
 - Gauge: 34 AWG (0.16 mm diameter)
 - Resistance: $34.5\text{ }\Omega/\text{meter}$
 - Operating Temperature: Up to 1150°C
 - Wire Length per Igniter: 0.15–0.20 m
 - Total Resistance per Igniter: 5–7 Ω
-

B Appendix B: PCB Design Details

B.1 PCB Specifications

- Dimensions: 65 mm $\times$ 65 mm
- Layers: 2 (top and bottom copper)
- Copper Weight: 2 oz (70 μ m)
- Material: FR-4, 1.6 mm thickness
- Surface Finish: ENIG (Electroless Nickel Immersion Gold)
- Via Size: 0.4 mm drill, 0.8 mm pad
- Trace Width: 0.25 mm minimum, 1.0 mm for high-current paths

B.2 Power Distribution

- Input: 14.8V from 4S LiPo via XT60 connector
- 5V Rail: LM2596 buck converter, 3A capacity
- 3.3V Rail: AMS1117-3.3 LDO, 1A capacity
- Power Planes: Dedicated ground plane on bottom layer
- Decoupling: 100 nF ceramic capacitors on all ICs, 100 μ F electrolytic on power inputs

B.3 Circuit Protection

- Input Protection: Reverse polarity protection diode (Schottky, 40V, 5A)
 - Over-current: Self-resetting fuses (PTC) on 5V and 3.3V rails
-

- MOSFET Protection: Flyback diodes on inductive loads
- ESD Protection: TVS diodes on communication lines

C Appendix C: Test Data

C.1 Telemetry Range Testing

Testing conducted on open field with clear line of sight. Transmitter at ground level, receiver elevated 2 m. 1000 packets transmitted per test point.

Table 6: Extended Telemetry Test Results

Distance (km)	Packets Sent	Packets Received	Loss (%)	RSSI (dBm)
0.5	1000	1000	0.0	-45
1.0	1000	998	0.2	-58
2.0	1000	995	0.5	-72
3.0	1000	981	1.9	-84
4.0	1000	966	3.4	-92
5.0	1000	953	4.7	-98

C.2 PWM Ignition Testing

Testing conducted with 0.18 m nichrome wire ($6.2\ \Omega$), 14.8V LiPo battery, room temperature (22°C).

Table 7: PWM Duty Cycle Ignition Tests

Duty Cycle (%)	Ignition Time (s)	Temperature (°C)	Wire Status	Replicates
50	No ignition	620	Intact	3
60	No ignition	750	Intact	3
70	4.2 ± 0.3	875 ± 25	Intact	5
80	2.8 ± 0.2	950 ± 30	Intact	3
100 (DC)	0.8 ± 0.1	>1000	Failed	2

Note: 70% duty cycle selected as optimal balance between ignition reliability and wire reusability.

C.3 Voltage Monitoring Accuracy

Comparison between multimeter reference and ADC reading. 10 measurements per voltage point.

Table 8: Voltage Monitoring Calibration

Reference (V)	ADC Reading (V)	Error (V)	Error (%)
12.00	11.88	-0.12	-1.0
13.00	12.92	-0.08	-0.6
14.00	13.95	-0.05	-0.4
14.80	14.73	-0.07	-0.5
15.60	15.68	+0.08	+0.5
16.40	16.52	+0.12	+0.7

Maximum absolute error: ±0.12V (±0.8%). Meets ±1% accuracy requirement.

D Appendix D: Project Budget

Table 9: Project Budget Breakdown

Item	Qty	Unit Cost (KES)	Total (KES)	Status
ESP32-WROOM-32	2	850	1,700	Received
XBee-PRO 900HP	2	4,500	9,000	Received
4S LiPo Battery (2200 mAh)	3	3,200	9,600	Received
BMS Board (4S)	3	650	1,950	Received
IRLZ44N MOSFET	10	45	450	Received
Nichrome Wire (10 m)	1	800	800	Received
PCB Fabrication (5 pcs)	1	4,500	4,500	Fabrication
PCB Components (resistors, caps, etc.)	1	2,000	2,000	Partial
Aluminum Stock (6061-T6)	1	1,500	1,500	Pending
PETG Filament (1 kg)	1	1,800	1,800	Received
Black Powder (100 g)	1	500	500	Pending
Connectors (XT60, JST, etc.)	1	1,200	1,200	Received
Fasteners and Hardware	1	800	800	Partial
Miscellaneous	1	1,500	1,500	–
Total Estimated Cost:			37,300	–

E Appendix E: Software Architecture

E.1 FreeRTOS Task Structure

1. **Sensor Task (10 Hz):** Reads GPS, IMU, barometer. Updates shared state variables via mutex.
2. **Telemetry Task (5 Hz):** Formats telemetry packet, transmits via XBee UART, receives ground commands.
3. **State Machine Task (20 Hz):** Manages flight phases, triggers deployment events, implements safety checks.
4. **Deployment Task (Event-driven):** Executes deployment sequence when triggered by state machine. Activates PWM, monitors current.
5. **Ignition Task (50 Hz):** Real-time PWM control with closed-loop current monitoring and thermal protection.
6. **Logging Task (10 Hz):** Writes sensor data and events to SD card in CSV format.
7. **Monitoring Task (1 Hz):** Checks battery voltage, igniter continuity, telemetry link health. Sets launch inhibit flags.

E.2 Telemetry Packet Format

Packet size: 64 bytes, transmitted at 5 Hz (320 bytes/s, well within XBee 10 kbps minimum).

Table 10: Telemetry Packet Structure

Field	Type	Bytes	Range/Notes
Header	uint16	2	0xAA55 sync pattern
Timestamp	uint32	4	Milliseconds since boot
GPS Latitude	float	4	Decimal degrees
GPS Longitude	float	4	Decimal degrees
GPS Altitude	float	4	Meters MSL
Barometric Altitude	float	4	Meters AGL
Velocity (vertical)	float	4	m/s
Acceleration X	int16	2	$\text{m/s}^2 \times 100$
Acceleration Y	int16	2	$\text{m/s}^2 \times 100$
Acceleration Z	int16	2	$\text{m/s}^2 \times 100$
Battery Voltage	uint16	2	mV
State	uint8	1	Flight state enum
Igniter Status	uint8	1	Bitfield: continuity, fired
Deployment Flags	uint8	1	Bitfield: drogue, main
Error Flags	uint8	1	Bitfield: sensor, battery, etc.
RSSI	int8	1	XBee RSSI
CRC	uint16	2	CRC-16 checksum
Total:		45	Padded to 64 bytes

E.3 Deployment Sequence Flowchart

1. **Trigger Condition Detected:** Apogee or main altitude threshold
2. **Check Safety Conditions:** Minimum altitude, minimum time, no previous deployment
3. **Safety Checks Passed?**
 - No: Abort, log error

- Yes: Continue
4. **Check Igniter Continuity:** Read continuity status
 5. **Continuity Good?**
 - No: Abort, log error, trigger backup timer
 - Yes: Continue
 6. **Activate PWM:** Start at 10% duty cycle
 7. **Ramp PWM:** Linear ramp to 70% over 2 seconds
 8. **Hold PWM:** Maintain 70% for 3 seconds
 9. **Monitor Current:** Verify expected current draw
 10. **Current Normal?**
 - No: Log warning, continue
 - Yes: Continue
 11. **Disable PWM:** Turn off after ignition window
 12. **Mark Deployment Complete:** Set deployment flag, prevent re-trigger
 13. **Log Event:** Record deployment event with timestamp
 14. **Continue Telemetry:** Monitor descent

F Appendix F: Safety Protocols

F.1 Pre-Launch Safety Checklist

1. Battery fully charged and voltage $\geq 14.5\text{V}$
 2. All igniters installed and continuity verified
-

3. Telemetry link established with $\pm 5\%$ packet loss
4. GPS lock acquired (≥ 6 satellites)
5. Parachutes properly packed with shock cord attached
6. Explosion caps secured with O-rings installed
7. All airframe connections tight and verified
8. Launch site clear of personnel (≥ 100 m radius)
9. Ground station operational with clear data display
10. Weather conditions acceptable (wind ≤ 10 m/s, no precipitation)
11. Launch inhibit released only when ready for ignition
12. Recovery team briefed and positioned

F.2 Launch Inhibit Conditions

System will refuse to arm igniters if ANY of the following conditions exist:

- Battery voltage ≤ 14.0 V
 - Igniter continuity not detected (drogue or main)
 - Telemetry link not established
 - GPS lock not acquired
 - Barometer failure detected
 - IMU failure detected
 - Ground station command inhibit active
-

F.3 Emergency Procedures

Lost Telemetry Link:

- System continues autonomous operation using onboard sensors
- Backup timer-based deployment activates if altitude data lost
- SD card logging continues for post-flight analysis
- GPS coordinates in last telemetry packet guide recovery

Igniter Failure:

- Backup timer-based deployment triggers at $T + [\text{time}]$ or altitude threshold
- Manual deployment via ground command (if telemetry available)
- Parachute extraction via motor ejection (if available on airframe)

Premature Deployment:

- State machine guard prevents deployment below minimum altitude (100 m AGL)
 - State machine guard prevents deployment before minimum time (10 seconds from liftoff)
 - Dual-switch safety requires both software enable and hardware enable
-